

Causal Emergence 2.0 Is Not Invariant Under Adjoining an Independent System

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Abstract

Causal Emergence 2.0 (CE 2.0) measures how much a system's coarse-grained scales add to its causal workings, using the causal primitives (CP) of determinism and specificity. This note shows that the measure is not invariant under adjoining an independent system.

For independent Markov chains A and B, the CP of the composite $A \otimes B$ is the log-size-weighted average of CP(A) and CP(B), under any intervention distribution that factorizes over the two components. The partition that coarse-grains away B is exactly lumpable and recovers CP(A). CE 2.0 therefore reports emergence for the composite whenever $CP(A) > CP(B)$, including when A alone shows none. The reported amount approaches $CP(A) - CP(B)$ as B is enlarged. Jansma and Hoel's full-lattice "emergent hierarchy" is produced the same way: a deterministic two-state system joined to four states of noise yields 196 emergent scales, all of which coarse-grain the noise.

CE 1.0 does not show the effect, because effective information adds across independent components. The difference traces to the size normalization that gives CE 2.0 its scale invariance. All results are computed with the authors' toolkit.

1. Setup

Causal Emergence 2.0 (CE 2.0) quantifies how much a system's coarse-grained descriptions add to its causal workings beyond its microscale (Hoel 2026; Jansma & Hoel 2025). The analysis begins from a system given as a Markov chain. Jansma and Hoel state the standing conditions explicitly: the analysis is confined "to causal models that are already known in their entirety, and are represented as Markov chains," each described by a given TPM; the microscale is assumed to be "well-defined, in that it forms a specific TPM," and "there are no hidden confounders" (Jansma & Hoel 2025, §2 and SI S1). This note works entirely within those conditions.

Notation. Let T be a transition probability matrix (TPM) on n states, with rows $T_{\cdot c} = T(\cdot | c)$, and let p be an intervention distribution over causes. Write $L = \log_2 n$ and H for Shannon entropy in bits.

Following Hoel (2026) and Jansma and Hoel (2025, eqs. 1–10), define:

- determinism: $\det(T; p) = 1 - \sum_c p(c) H(T_{\cdot c}) / L$
- specificity: $\text{spec}(T; p) = H(E) / L$, where $E = \sum_c p(c) T_{\cdot c}$ is the effect distribution

- causal primitives: $CP(T; p) = \det(T; p) + \text{spec}(T; p) - 1$

This is the determinism–specificity form of CP, which Jansma and Hoel (2025) use throughout. It equals the "effectiveness" of CE 1.0 (Hoel, Albantakis & Tononi 2013) and is the default mode of the authors' pymergence toolkit. Hoel (2026) also reports a sufficiency–necessity form; §3 treats it separately. Hoel recommends a uniform p at each scale and reports that emergence also appears under other choices, including the observed distribution (Hoel 2026, "An axiomatic grounding").

A **scale** is a partition π of the n states. Its macro TPM T^π is obtained by averaging the rows of each block's member states and summing over each target block (Hoel et al. 2013; Klein & Hoel 2020). Two operations on scales matter here:

- **Dynamical consistency.** Hoel (2026, Note S2) keeps a macroscale only if random walkers started from every state follow the same coarse-grained trajectory at both scales over five steps, measured by KL divergence. Jansma and Hoel (2025, §4) drop this filter.
- **Emergence scores.** The degree of causal emergence is the total gain in CP along a micro-to-macro path, taken to the macroscale of highest CP (Hoel 2026). Along any single path it telescopes to $CE(T) = \max_\pi CP(T^\pi) - CP(T)$, the maximum running over admissible scales. The full-lattice variant assigns each scale $\Delta CP(\pi) = CP(T^\pi) - \max_{\pi'} CP(T^{\pi'})$, the maximum running over all strictly finer π' . The scales with positive ΔCP form the "emergent hierarchy" (Jansma & Hoel 2025, §3.1).

The construction. Let T_A on n_A states and T_B on n_B states be any two TPMs. Their independent composition is the Kronecker product $T = T_A \otimes T_B$ on the $n_A \cdot n_B$ joint states (a, b) :

$$T((a', b') | (a, b)) = T_A(a' | a) \cdot T_B(b' | b)$$

Neither component influences the other. The composite is a fully specified Markov chain whose state at t fixes its distribution at $t+1$, and it contains no unmodeled variable. It therefore meets every condition quoted above. Write $L_A = \log_2 n_A$ and $L_B = \log_2 n_B$, and define the weights

$$w_A = L_A / (L_A + L_B), \quad w_B = L_B / (L_A + L_B), \quad \text{so that } w_A + w_B = 1.$$

2. Proposition 1: the causal primitives of an independent composition are a weighted average

Proposition 1. For any intervention distributions p_A and p_B , let $p = p_A \otimes p_B$. Then

$$\begin{aligned} \det(T_A \otimes T_B; p) &= w_A \cdot \det(T_A; p_A) + w_B \cdot \det(T_B; p_B) \\ \text{spec}(T_A \otimes T_B; p) &= w_A \cdot \text{spec}(T_A; p_A) + w_B \cdot \text{spec}(T_B; p_B) \\ CP(T_A \otimes T_B; p) &= w_A \cdot CP(T_A; p_A) + w_B \cdot CP(T_B; p_B) \end{aligned}$$

Proof.

1. *Determinism.* Each row of the composite is a product distribution, $T_{(a,b)} = T_{A,a} \otimes T_{B,b}$, so its entropy is additive: $H(T_{(a,b)}) = H(T_{A,a}) + H(T_{B,b})$. Averaging under $p = p_A \otimes p_B$

gives $\sum p(a,b) H(T_{(a,b)}) = h_A + h_B$, where $h_A = \sum_a p_A(a) H(T_A, a) = L_A[1 - \det(T_A; p_A)]$ and h_B is defined likewise. Because $\log_2(n_A \cdot n_B) = L_A + L_B$, $\det(T; p) = 1 - (h_A + h_B)/(L_A + L_B)$, which rearranges to the weighted average.

2. *Specificity*. The effect distribution factorizes: $E = \sum p(a,b) T_{(a,b)} = E_A \otimes E_B$. So $H(E) = H(E_A) + H(E_B) = L_A \cdot \text{spec}(T_A; p_A) + L_B \cdot \text{spec}(T_B; p_B)$. Dividing by $L_A + L_B$ gives the weighted average.
3. *CP*. Add the two identities and subtract $1 = w_A + w_B$. ■

The identity is exact for every product intervention distribution. That class includes the uniform distribution at the joint microscale (uniform on $n_A \cdot n_B$ states is the product of the uniforms) and the observed stationary distribution whenever each component has a unique stationary distribution, since the composite's stationary distribution is then the product of the components'. Across 300 randomly generated pairs with random product intervention distributions, the identity holds in the authors' toolkit to within 4×10^{-16} .

3. Proposition 2 and corollary: projecting onto a component

Let π_A be the partition of the joint states that groups (a, b) with (a, b') for all b, b' . Its blocks correspond one-to-one with the states of A .

Proposition 2.

- (i) $T_A \otimes T_B$ is exactly lumpable with respect to π_A , and its macro TPM is T_A .
- (ii) π_A is dynamically consistent at every horizon.
- (iii) CP of the macroscale π_A equals CP(T_A) under the uniform distribution at that scale, and under p_A when the joint intervention distribution is $p_A \otimes p_B$.

Proof.

- (i) For any joint state (a, b) and any target block $\{(a', \cdot)\} : \sum_{b'} T((a', b') | (a, b)) = T_A(a' | a) \cdot \sum_{b'} T_B(b' | b) = T_A(a' | a)$. This does not depend on b . Every member state of a block therefore has the same block-transition probabilities, so any within-block averaging, uniform or otherwise, yields T_A .
- (ii) Under exact lumpability, the projection of the micro random-walker distribution equals the macro random-walker distribution at every step, so the KL criterion of Hoel (2026, Note S2) is zero at every horizon.
- (iii) Follows from (i). The uniform distribution on the blocks is the uniform distribution on the states of A , and coarse-graining $p_A \otimes p_B$ onto π_A returns p_A . ■

Corollary. Under the uniform intervention distribution at each scale, or under the observed distribution at each scale, the causal emergence reported for the composite satisfies

$$CE(T_A \otimes T_B) \geq CP(T_A) - CP(T_A \otimes T_B) = w_B \cdot [CP(T_A) - CP(T_B)].$$

1. *Positivity.* The bound is strictly positive whenever $CP(T_A) > CP(T_B)$. This holds whatever emergence T_A shows on its own. In particular, when T_A has no emergence (no partition of A raises its CP), CE 2.0 still reports emergence for the composite, and it locates that emergence at the scale π_A , which is T_A itself.
2. *Dialing.* Fix T_A and let B be uniform noise on n_B states, so $CP(T_B) = 0$. As n_B grows, $w_B \rightarrow 1$ and the bound approaches $CP(T_A)$. The emergence attributed to the composite can therefore be raised toward $CP(T_A)$ by enlarging an appended system that does nothing.
3. *Symmetry.* If $CP(T_B) > CP(T_A)$, the same argument applies with A and B exchanged.
4. *Scope of the bound.* The corollary gives a lower bound. $CE(T_A \otimes T_B)$ may exceed it if some partition of the joint states achieves CP above $CP(T_A)$.

Remark on the sufficiency–necessity form. Hoel (2026) also computes CP as average sufficiency plus average necessity, minus 1. Proposition 1 does not carry over, because average sufficiency is multiplicative under independent composition. The qualitative result survives:

- For T_A the 2-state identity and T_B uniform on n_B states, this form reports a gain at π_A of 0.667, 0.857, 0.933 and 0.968 for $n_B = 2, 4, 8$ and 16.
- Across 2,000 random pairs, the composite's CP falls strictly below the better component's CP in 1,997 cases. The three exceptions occur where both components are close to $CP = 0$.

The closed-form statements in this note are made for the determinism–specificity form only.

4. Worked example

Let T_A be the 2-state identity: each state returns to itself with probability 1. Let T_B be uniform noise on n_B states: every row is uniform. The two never interact.

A on its own. $CP(T_A) = 1$, since determinism and specificity are both maximal. No partition of A can raise its CP, so A shows no causal emergence.

B on its own. $CP(T_B) = 0$ at every scale.

The composite at $n_B = 2$. $A \otimes B$ has four joint states: $0 = (a_0, b_0)$, $1 = (a_0, b_1)$, $2 = (a_1, b_0)$, $3 = (a_1, b_1)$. By Proposition 1 its microscale CP is $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = 0.5$. The hand check is short:

- From any state, A 's component stays fixed and B 's is a fair coin. The row entropy is therefore 1 bit out of a possible 2, which gives determinism 0.5.
- Under uniform interventions every joint state is an equally likely effect, so specificity is 1.

The micro-to-macro path the toolkit selects, which is the longest path to the macroscale of highest CP, runs:

Scale	CP	Gain
(0)(1)(2)(3), the joint microscale	0.500	
(0)(1)(2,3)	0.579	0.079
(0,1)(2,3) = π_A	1.000	0.421

The total gain is 0.500, which equals CE. Both macroscales pass Hoel's (2026) consistency criterion. The endpoint π_A is exactly "the state of A." The intermediate scale merges the two B-states only when A is in a_1 . Computed over the two gains, emergent complexity along this path is 0.63 bits. Taken on its own, A has none.

Dialing. Holding T_A fixed and enlarging the uniform B gives the following. Because π_A attains the maximum possible CP of 1, the corollary's bound is met exactly here.

n_B	CP(A $\otimes$ B)	CE reported
2	0.500	0.500
4	0.333	0.667
8	0.250	0.750
16	0.200	0.800
64	0.143	0.857

The limit is 1, the largest value CE 2.0 can report. Nothing in A changes from row to row; only the appended system grows.

5. The emergent hierarchy is produced the same way

Jansma and Hoel's (2025) full-lattice analysis turns a system's scales into an "emergent hierarchy": the scales that add CP over every finer scale beneath them. For A alone that hierarchy is empty.

- **$n_B = 2$.** A $\otimes$ B has three emergent scales with $\Delta CP > 0.01$: π_A (ΔCP 0.421) and the two partial groupings (0)(1)(2,3) and (0,1)(2)(3) (ΔCP 0.079 each).
- **$n_B = 4$.** The eight-state composite has 4,140 partitions. Of these, 196 have $\Delta CP > 0.01$, and 64 exceed 0.05. The 196 are spread across every level from 7 blocks down to 2:

Blocks	7	6	5	4	3	2
Emergent scales	12	36	84	49	14	1

All 196 pass the consistency criterion. None of them groups joint states that differ in A. Each one merges B-states within fixed values of A, and the single two-block scale is π_A itself. There are 6,480 maximal micro-to-macro paths ending at π_A , each six steps long. Every one of them totals 0.667, and emergent complexity along them ranges from 1.43 to 1.73 bits.

What the full-lattice analysis presents as a mesoscale-rich hierarchy is, in this case, the lattice of partial marginalizations of a noise process. That process was appended to a system that has no hierarchy of its own. Two cautions for reading these figures:

- **Not additive.** The ΔCP values do not sum to anything conserved. Their total here is 9.74. Jansma and Hoel note that "it is generally not the case that the sum of all ΔCP values adds up to the maximum attainable CP of 1."
- **Not classified.** This note does not assign the $n_B = 4$ hierarchy a place in Jansma and Hoel's taxonomy (top-heavy, bottom-heavy, mesoscale), because that requires their path-entropy and row-negentropy measures. The point is only that the hierarchy exists and that every scale in it comes from B.

6. Proposition 3: effective information adds, so CE 1.0 shows no such gain

In CE 1.0, emergence is measured by effective information. For a TPM T with intervention distribution p :

$$EI(T; p) = I(C; E) = H(E) - \sum_c p(c) H(T_c) = L \cdot CP(T; p)$$

Proposition 3. For $T = T_A \otimes T_B$ and $p = p_A \otimes p_B$:

$$EI(T; p) = EI(T_A; p_A) + EI(T_B; p_B)$$

Proof. Multiply the CP identity of Proposition 1 by $L_A + L_B$. ■

Because $EI(T_B; p_B) \geq 0$, the projection π_A can never carry more effective information than the joint microscale. The mechanism of §3 therefore cannot operate in CE 1.0. In the worked example, the maximum EI over all partitions equals the microscale EI of 1.000 bit for both $n_B = 2$ and $n_B = 4$, so CE 1.0 reports no emergence. The general claim in this section is limited to π_A . It is not a claim about every partition of every composite.

The source of the difference. The difference comes from the factor $L = \log_2 n$. Hoel (2026) removes this "size term" because it "prevents CE 1.0 from being able to analyze the multiscale structure clearly since it is not scale invariant." Dividing EI by L makes scales of different sizes comparable. The same division turns additivity across independent components into a size-weighted average.

Scale invariance, and protection against the mechanism of §3, are therefore traded against each other through one normalization. CE 1.0 has the protection and lacks scale invariance; CE 2.0 has scale

invariance and lacks the protection. The note takes no position on which trade is better, and CE 1.0 has limitations of its own (Eberhardt & Lee 2022; Hoel 2026).

Aaronson's normalization. The same factor bears on Aaronson's (2017) objection to CE 1.0. Aaronson argued that the macro intervention distribution should be obtained by marginalizing the micro one. Under that requirement, the data processing inequality ensures that mutual information cannot rise under coarse-graining. Hoel (2026) reports that CE 2.0 still detects emergence when the observed distribution is used at both scales, which satisfies the requirement. Any such gain is therefore supplied by the division by L . That division is also the source of the effect in §3, and Propositions 1 and 2 hold under Aaronson's requirement, since the marginal of $p_A \otimes p_B$ on π_A is p_A .

7. Robustness

- **Intervention distribution.** Proposition 1 holds for every product distribution. That covers the uniform distribution, which Hoel recommends at each scale, and the observed stationary distribution wherever the components' stationary distributions are unique. For an independent composite, product distributions are also the only choice that respects the system's structure: a non-product intervention distribution would impose a dependence between A and B that the dynamics do not contain. Hoel (2026) argues that CE 2.0 detects emergence "across diverse background assumptions." That argument does not bear on the present result, which holds across the same assumptions.
- **Consistency filter.** π_A is exactly lumpable, so the result holds with the filter (Hoel 2026) or without it (Jansma & Hoel 2025). Every emergent scale reported in §5 passes the filter.
- **Scoring scheme.** Both the path-wise CE of Hoel (2026) and the full-lattice ΔCP of Jansma and Hoel (2025) register the effect.
- **Form of CP.** The closed forms apply to the determinism–specificity form. The sufficiency–necessity form shows the same effect numerically, with larger gains in the worked example (§3, remark).
- **Implementation.** All values were computed with the authors' pymergence toolkit at commit 6e5cb45 (26 September 2025). The toolkit builds macro TPMs by averaging each member state's block sums, and tests consistency with delta-started random walkers over five steps, using a tolerance of 10^{-2} . Every projection reported here is exactly lumpable and scores exactly zero. Both procedures match the procedures described in Hoel (2026). The script and its full output accompany this note.

8. Relation to prior work

Elementary status. Both propositions are elementary. Proposition 1 is the additivity of entropy over independent variables (Cover & Thomas 2006). Proposition 2(i) is a standard case of exact lumpability (Kemeny & Snell 1960). What this note contributes is their consequence for CE 2.0's emergence scores

and emergent hierarchies.

Eberhardt and Lee (2022). The closest critique is theirs, and it differs from the present result in three respects:

1. *Operation.* Their Section 4 shows that abstraction and marginalization do not commute in CE 1.0. The marginalization there is over time steps, comparing the abstraction of T^2 with the square of the abstraction of T , or over the internal link of a mechanism chain. The present result concerns which state variables the modeled system contains.
2. *Diagnosis.* Eberhardt and Lee trace the artifacts they identify to "the introduction of a maximum entropy distribution that is extraneous to the system under investigation." The present result holds for every product intervention distribution, including the observed one (§7).
3. *Target.* Jansma and Hoel (2025) state that CE 2.0 "negates previous mathematical criticisms of the theory," citing Eberhardt and Lee. That claim, whatever its scope for their critique, does not reach this one.

The two results do share a conclusion. Eberhardt and Lee conclude that the account describes "features that crucially depend on the rate of measurement (if we think in terms of time series) or on the input–output points (if we think of $X \rightarrow Y$ as a mechanism) chosen by the investigator." The present result adds a third investigator-chosen input: the boundary of the modeled system.

A minimal desideratum. Eberhardt and Lee frame commutativity as a desideratum that any account of multi-scale causation should satisfy. The present result motivates a weaker one: *if neither of two causally independent systems exhibits causal emergence, their composite should not*. CE 2.0 violates it in the worked example, where A shows no emergence, B shows none at any nontrivial scale, and the composite shows $CE = 0.5$. CE 1.0 satisfies it in the same example. The note does not claim that CE 1.0 satisfies it in general.

The standing assumptions of CE 2.0. Jansma and Hoel's conditions ensure the model is complete: nothing relevant is hidden, and the TPM is known in its entirety. The present result shows that the conditions leave the model's minimality unconstrained. Nothing in them prevents the declared state space from including components that play no part in the organization whose emergence is being measured, and CE 2.0's outputs depend on that as much as on completeness.

Other work. Aaronson (2017) and the discussion that followed concern CE 1.0's intervention distribution. Two comments in that discussion bear on the present result.

- A commenter signing as "Will" (comment #104) proposed attaching to a system "a cup of hot tea and a jug of cold milk" that "has no connection to the output whatsoever." The attachment is rigged to equalize the number of microstates within each macrostate, and this removes CE 1.0's emergence. That construction couples the added component to the macrostate, and it runs in the opposite direction to the present result.
- A commenter signing as "Michael" (comment #85) proposed normalizing mutual information

by the logarithm of the number of interventions, observing that "inventing meaningless micro-states will be strictly harmful." Hoel replied that this is close to the effectiveness measure of his theory (comment #92). That is the mechanism of Proposition 1, read in the discussion as a virtue of the measure.

Dewhurst (2021) asks whether EI-based emergence is causal and whether it is emergent, and concludes that it is at most epistemic. His paper does not address how the modeled system is composed. The author is not aware of prior work stating the non-invariance of CE 2.0 under independent composition, or its consequences for emergence scores and emergent hierarchies.

The origin of effective information. Effective information was introduced to quantify causal interaction between two complementary parts A and B of a system S, by replacing A's outputs with independent noise and measuring the mutual information this produces in B (Tononi & Sporns 2003, as presented in Dewhurst 2021). Applied across the bipartition {A, B} of the composite in §4, that original measure returns zero in both directions. The measure's own lineage therefore contains a test that registers $A \otimes B$ as two non-interacting parts. CE 1.0 and CE 2.0 apply the measure instead to transitions between states of the declared whole, where no such test is made. Reintroducing a check of this kind before computing CE would amount to deciding which parts belong together before measuring emergence among the states of the whole.

9. Scope

The propositions concern exact independence. One part of the argument extends further. Proposition 2 requires only that A evolve autonomously: A may influence B, provided B does not influence A. Under that condition, π_A remains exactly lumpable with macro TPM T_A , and CE 2.0 reports the gain $CP(T_A) - CP(T)$ whether B is independent noise or a component driven by A.

In a two-state example where B copies A's state with probability q and otherwise flips a coin, the reported emergence is 0.5 for $q = 0, 0.5$ and 0.9 . Across that range B goes from unrelated to closely tracking A, and CE 2.0's output does not register the difference. Proposition 1 does not extend to driven cases, and coupling in which B influences A is not treated here.

The partition CE 2.0 selects in these cases, π_A , is the sensible one: it recovers A exactly. The dependence on the boundary lies in the emergence score and in the hierarchy of scales, which attribute to the composite a gain that belongs to A's own organization.

A defender may reply that a measure which rewards removing meaningless degrees of freedom is working as intended. The reply concedes the mechanism and leaves the attribution in place. CE 2.0 records the removal of B as causal emergence of the composite, and it enters the resulting scales into the composite's emergent hierarchy. Nothing emerges in A, and A's organization was fully present before B was included. The minimal desideratum of §8 marks the difference between rewarding a coarse-graining and calling its result emergence.

This note does not claim that emergence is illusory, that CE 2.0 fails at its stated purpose, or that any

published CE analysis contains an adjoined independent component. Whether a given application is affected requires examining how its state space was declared.

10. Interpretation

Eberhardt and Lee (2022) offer two readings of results of this kind:

"To the extent that one reads Hoel's account as a description of objective reality, our results show that the description is not objective, but contains significant modeling artifacts. If instead one interprets Hoel's account as merely epistemic, then the results point to an arbitrariness in the investigator's model that is not well justified and that has misleading consequences."

Dewhurst (2021) argues on independent grounds that the second reading is the one EI-based emergence can support. The same fork applies here.

- *On the objective reading*, part of the emergence CE 2.0 reports for a composite is a product of where the model's boundary was drawn.
- *On the epistemic reading*, the boundary is a free parameter of the analysis that CE 2.0 neither sets nor constrains.

On either reading, deciding which states belong to the system comes before the measure can be interpreted. CE 2.0 can recover causally powerful organization from an overinclusive model. It cannot determine, from its own outputs, whether that organization emerged within the modeled whole or belonged all along to a component that should have been modeled on its own.

Criteria for that prior question, what warrants treating a set of states as one system, are developed elsewhere (Thomas 2026a; Thomas 2026b).

Code and data availability

The script `two_systems_as_one.py` and its full output accompany this record. It reproduces every value reported here using `pymergence` (Jansma & Hoel 2025b) at commit 6e5cb45 (26 September 2025), <https://github.com/EI-research-group/pymergence>. Runtime is about two minutes.

Methodology

Computations and an initial draft were prepared with the assistance of Claude (Anthropic). The author reproduced the reported values independently and is responsible for the content.

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